Initial Project Idea

ECE 382V, Fall 2026

Are LLM Written Unit Tests Flaky?

We want to find out whether unit tests written by LLMs are flaky, meaning they can pass and fail on the same code without anything changing. Tools like Copilot and research tools like TestPilot usually generate a test, run it once, and keep it if it passes, so a flaky test can slip in and only start failing later in CI. Most of the flaky test studies we have read look at tests written by people, so it is not clear how LLM written tests compare. Our idea is to take a set of small Python libraries and have a few LLM setups write tests for them, including an agent that can run and fix its own tests and models that write tests without ever running them. We would then rerun those suites and the human written tests that come with each library many times while changing the test order, the hash seed, the time zone, and the CPU load, and count a test as flaky if it both passes and fails. To evaluate this we would compare flaky rates between the LLM and human tests, label the root cause of each flaky test with the Luo et al. categories, and check whether simple patterns like real sleeps or unfrozen clock reads predict which tests flip. If the LLM tests turn out to be riskier, we want to try a fix such as a prompt that warns about these patterns or rerunning new tests a few times before keeping them. A small pilot on a couple of libraries already found an LLM written test that only fails after 10 PM. The main thing we are unsure about is that flaky tests are rare, so we may need a lot of reruns or a slower machine to see real differences.

Team

Shaunak Kapur (sk62288) and Manas Pathak (map7848)